Scarab ransomware spreading
An email campaign from Necurs is spreading new ransomware at over 2 million emails per hour. **http://dgit.in/Scambot**

Airbnb for money laundering
Russian scammers are leveraging the service to launder dirty cash from stolen credit cards with the help of corrupt hosts. **http://dgit.in/Airlaunder**

Defence Against the Dark Asteroids

Are we fortified enough to survive a potentially devastating asteroid headed towards Earth?

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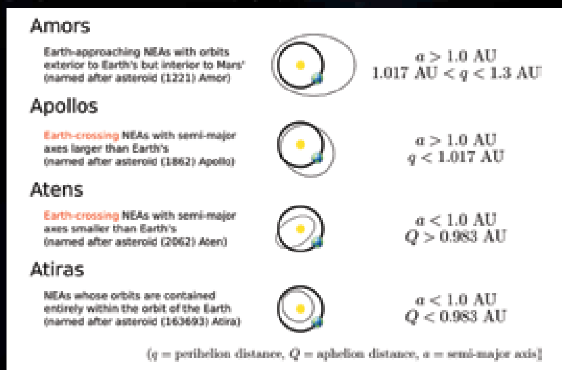
We can't be certain which meal or train ride would be our last. But if there are space rocks drifting towards us from millions of kilometres away, we can certainly make efforts to identify them before they wipe us out. There are an infinite number of rocks floating in space, some of which are orbiting the sun. If our planet's orbital path happens to cross their's, we could have a problem (putting it mildly). However not all rocks can cause catastrophic damage – there's a set of criteria that the bodies need to fulfil before being deemed as dangerous. Bodies the size of around 25 metres across are more likely to burn up before coming close enough to the surface to cause damage. Objects between 25 metres and one kilometer will most likely cause damage around the area of impact. Any object bigger than that or around one to two kilometres across might have global effects. Near-Earth Objects (NEOs) are asteroids or comets that have a perihelion (nearest point of an object with the Earth in an elliptical orbit) distance of less than 1.3 au (Astronomical Unit which is approximately the average distance between the Earth and the Sun, i.e., about 150 billion metres). NEOs are further divided into Near-Earth Comets (NECs) and Near-Earth Asteroids (NEAs). NECs are comets which have an orbital period (time taken by the comet to complete one orbit around the Earth) of less than 200 years. NEAs are further classified into three groups – Atira, Apollo and Amor – based on their distances explained in the image below.



Credit: Center for Near Earth Object Studies | NASA

Different groups of Near-Earth Asteroids

NEAs are redefined into Potentially Hazardous Asteroids (PHAs) based on whether they are capable of making potentially harmful close approaches to our planet. The minimum distance is around 0.05 au or roughly 7,480,000 km and the minimum size of the asteroid is around 140m. Considering the infinite number of objects floating around space and orbiting the sun, you'd wonder whether there are asteroids already on their way to hit Earth. According to a report by NASA's Near-Earth Object Wide-Field Infrared Survey Explorer (NEOWISE) in 2012, they identified about 4,700 PHAs, plus or minus 1,500, and their diameters were larger than 100 metres. And it is estimated that only 20 to 30 percent of these objects have been discovered. To be completely sure about objects which are on a collision course towards the third rock from the Sun, we need to scour the expanse of the observable universe. Once labelled as potentially dangerous, such bodies are marked and continuously monitored.

Monitoring hazardous asteroids

When it comes to monitoring for threats from space rocks, we are only interested in the potentially hazardous objects. Predicting or rather calculating their collision course is vital since surviving the impact would be on top priority if the object happens to be headed towards Earth. To be on top of the game, NASA pushed its asteroid-hunting efforts under the Planetary Defense Coordination Office (PDCO). Their primary responsibilities include early detection of PHAs, tracking and characterising them and issuing warnings about possible impacts. If detected, they are supposed to provide accurate information about the PHAs periodically and coordinate with the US Government to plan while dealing with an actual threat. NASA has set up automatic impact detection systems where Scout is used to locate and alert us about incoming threats from space rocks. The other program is called Sentry which looks at bigger and potentially dangerous objects. Apart from this, astronomers across the world, whether amateur or professional are contributing to the discovery of asteroids

Defending ourselves from an asteroid impact

Nihilists aside, most of us on Earth want to live longer. Coming up with